2.5 Continuity

Continuity at a Point

To understand continuity, it helps to consider a function like that in Figure 2.35, whose limits we investigated in Example 2 in the last section.

EXAMPLE 1 At which numbers does the function f in Figure 2.35 appear to be not continuous? Explain why. What occurs at other numbers in the domain?

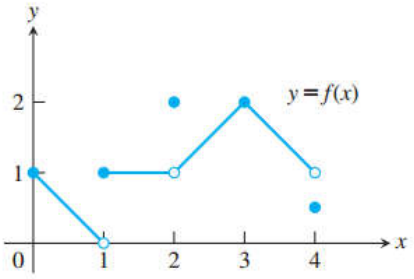


FIGURE 2.35 The function is not continuous at $x = 1$, $x = 2$, and $x = 4$ (Example 1).

DEFINITIONS Let c be a real number that is either an interior point or an endpoint of an interval in the domain of f.

The function f is **continuous at c** if

$$\lim_{x \to c} f(x) = f(c).$$

The function f is **right-continuous at c (or continuous from the right)** if

$$\lim_{x \to c^+} f(x) = f(c).$$

The function f is **left-continuous at c (or continuous from the left)** if

$$\lim_{x \to c^-} f(x) = f(c).$$

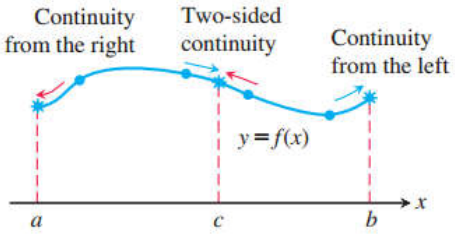


FIGURE 2.36 Continuity at points a, b, and c.

EXAMPLE 2 The function $f(x) = \sqrt{4 - x^2}$ is continuous over its domain $[-2, 2]$ (Figure 2.37). It is right-continuous at $x = -2$, and left-continuous at $x = 2$. ■

EXAMPLE 3 The unit step function $U(x)$, graphed in Figure 2.38, is right-continuous at $x = 0$, but is neither left-continuous nor continuous there. It has a jump discontinuity at $x = 0$. ■

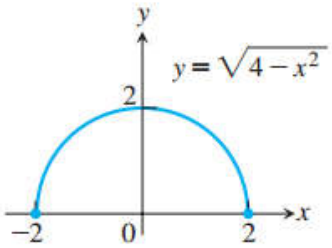


FIGURE 2.37 A function that is continuous over its domain (Example 2).

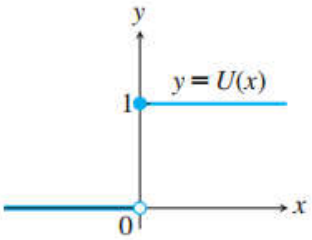


FIGURE 2.38 A function that has a jump discontinuity at the origin (Example 3).

Continuity Test

A function $f(x)$ is continuous at a point $x = c$ if and only if it meets the following three conditions.

1. $f(c)$ exists	(c lies in the domain of f).
2. $\lim_{x \to c} f(x)$ exists	(f has a limit as $x \to c$).
3. $\lim_{x \to c} f(x) = f(c)$	(the limit equals the function value).

For one-sided continuity, the limits in parts 2 and 3 of the test should be replaced by the appropriate one-sided limits.

of 2. The discontinuity in Figure 2.40c is **removable**. The function has a limit as $x \to 0$, and we can remove the discontinuity by setting $f(0)$ equal to this limit.

The discontinuities in Figure 2.40d through f are more serious: $\lim_{x \to 0} f(x)$ does not exist, and there is no way to improve the situation by appropriately defining f at 0. The step function in Figure 2.40d has a **jump discontinuity**: The one-sided limits exist but have different values. The function $f(x) = 1/x^2$ in Figure 2.40e has an **infinite discontinuity**. The function in Figure 2.40f has an **oscillating discontinuity**: It oscillates so much that its values approach each number in $[-1, 1]$ as $x \to 0$. Since it does not approach a single number, it does not have a limit as x approaches 0.

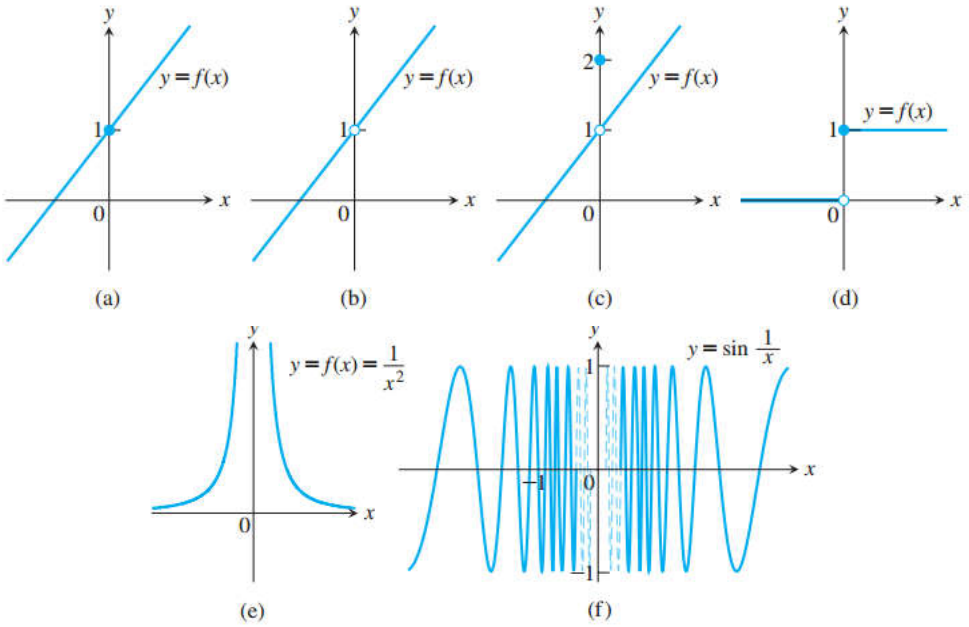


FIGURE 2.40 The function in (a) is continuous at $x = 0$; the functions in (b) through (f) are not.

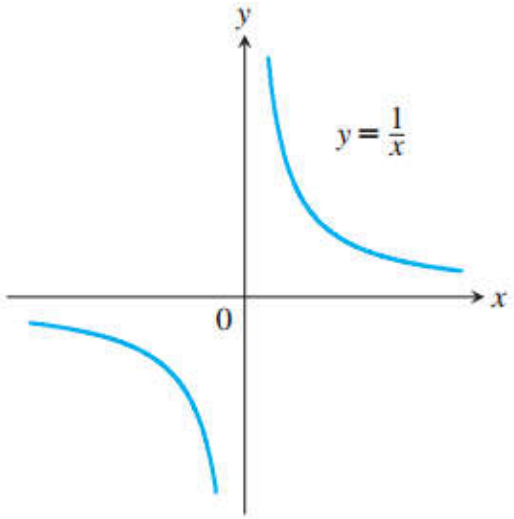


FIGURE 2.41 The function $f(x) = 1/x$ is continuous over its natural domain. It is not defined at the origin, so it is not continuous on any interval containing $x = 0$ (Example 5).

EXAMPLE 2 Where are each of the following functions discontinuous?

(a) $f(x) = \dfrac{x^2 - x - 2}{x - 2}$

(b) $f(x) = \begin{cases} \dfrac{1}{x^2} & \text{if } x \neq 0 \\ 1 & \text{if } x = 0 \end{cases}$

(c) $f(x) = \begin{cases} \dfrac{x^2 - x - 2}{x - 2} & \text{if } x \neq 2 \\ 1 & \text{if } x = 2 \end{cases}$

(d) $f(x) = [\![x]\!]$

THEOREM 8—Properties of Continuous Functions

If the functions f and g are continuous at $x = c$, then the following algebraic combinations are continuous at $x = c$.

1. *Sums:*	$f + g$
2. *Differences:*	$f - g$
3. *Constant multiples:*	$k \cdot f$, for any number k
4. *Products:*	$f \cdot g$
5. *Quotients:*	f/g, provided $g(c) \neq 0$
6. *Powers:*	f^n, n a positive integer
7. *Roots:*	$\sqrt[n]{f}$, provided it is defined on an interval containing c, where n is a positive integer

5 THEOREM

(a) Any polynomial is continuous everywhere; that is, it is continuous on $\mathbb{R} = (-\infty, \infty)$.

(b) Any rational function is continuous wherever it is defined; that is, it is continuous on its domain.

7 THEOREM The following types of functions are continuous at every number in their domains:

polynomials rational functions root functions

trigonometric functions inverse trigonometric functions

exponential functions logarithmic functions

8 THEOREM If f is continuous at b and $\lim_{x \to a} g(x) = b$, then $\lim_{x \to a} f(g(x)) = f(b)$.
In other words,

$$\lim_{x \to a} f(g(x)) = f\left(\lim_{x \to a} g(x)\right)$$

EXAMPLE 8 Evaluate $\lim_{x \to 1} \arcsin\left(\frac{1 - \sqrt{x}}{1 - x}\right)$.

THEOREM 10—Limits of Continuous Functions
If $\lim_{x \to c} f(x) = b$ and g is continuous at the point b, then

$$\lim_{x \to c} g(f(x)) = g(b).$$

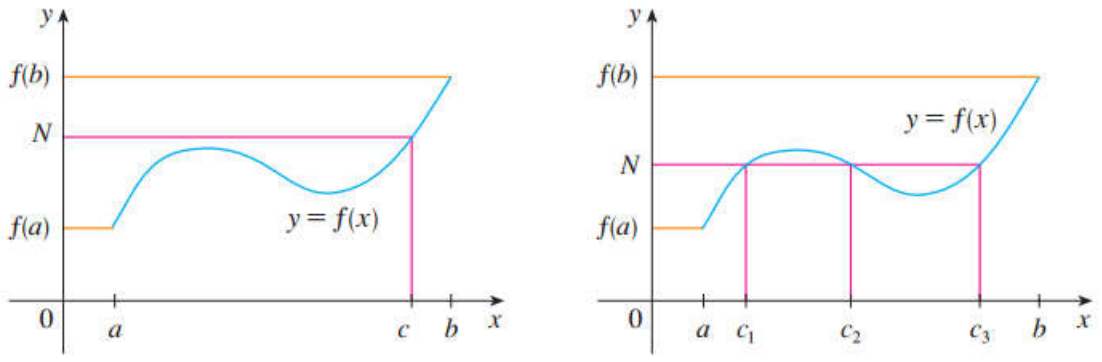
EXAMPLE 9 Where are the following functions continuous?
(a) $h(x) = \sin(x^2)$ (b) $F(x) = \ln(1 + \cos x)$

SOLUTION
(a) We have $h(x) = f(g(x))$, where

$$g(x) = x^2 \quad \text{and} \quad f(x) = \sin x$$

10 THE INTERMEDIATE VALUE THEOREM Suppose that f is continuous on the closed interval $[a, b]$ and let N be any number between $f(a)$ and $f(b)$, where $f(a) \neq f(b)$. Then there exists a number c in (a, b) such that $f(c) = N$.

The Intermediate Value Theorem states that a continuous function takes on every intermediate value between the function values $f(a)$ and $f(b)$. It is illustrated by Figure 8. Note that the value N can be taken on once [as in part (a)] or more than once [as in part (b)].



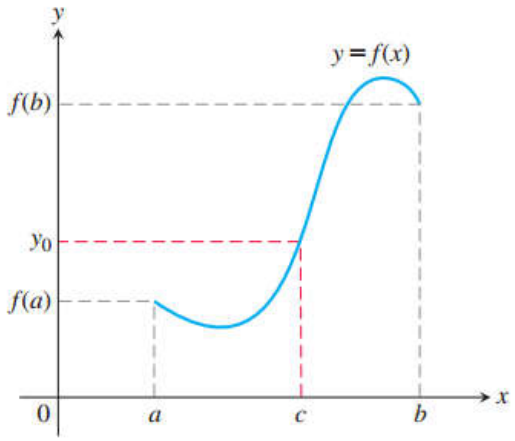
(a) (b)

Intermediate Value Theorem for Continuous Functions

A function is said to have the **Intermediate Value Property** if whenever it takes on two values, it also takes on all the values in between.

THEOREM 11—The Intermediate Value Theorem for Continuous Functions

If f is a continuous function on a closed interval $[a, b]$, and if y_0 is any value between $f(a)$ and $f(b)$, then $y_0 = f(c)$ for some c in $[a, b]$.



A Consequence for Graphing: Connectedness Theorem 11 implies that the graph of a function that is continuous on an interval cannot have any breaks over the interval. It will be **connected**—a single, unbroken curve. It will not have jumps such as the ones found in the graph of the greatest integer function (Figure 2.39), or separate branches as found in the graph of $1/x$ (Figure 2.41).

A Consequence for Root Finding We call a solution of the equation $f(x) = 0$ a **root** of the equation or **zero** of the function f. The Intermediate Value Theorem tells us that if f is continuous, then any interval on which f changes sign contains a zero of the function. Somewhere between a point where a continuous function is positive and a second point where it is negative, the function must be equal to zero.

In practical terms, when we see the graph of a continuous function cross the horizontal axis on a computer screen, we know it is not stepping across. There really is a point where the function's value is zero.

EXAMPLE 10 Show that there is a root of the equation

$$4x^3 - 6x^2 + 3x - 2 = 0$$

between 1 and 2.

Continuous Extension to a Point

More generally, a function (such as a rational function) may have a limit at a point where it is not defined. If $f(c)$ is not defined, but $\lim_{x \to c} f(x) = L$ exists, we can define a new function $F(x)$ by the rule

$$F(x) = \begin{cases} f(x), & \text{if } x \text{ is in the domain of } f \\ L, & \text{if } x = c. \end{cases}$$

EXAMPLE 12 Show that

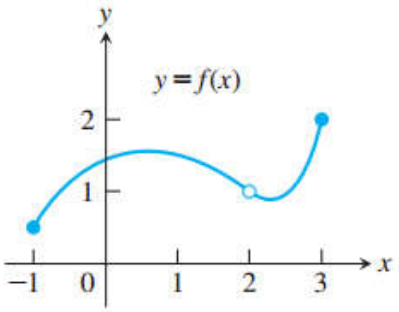
$$f(x) = \frac{x^2 + x - 6}{x^2 - 4}, \quad x \neq 2$$

has a continuous extension to $x = 2$, and find that extension.

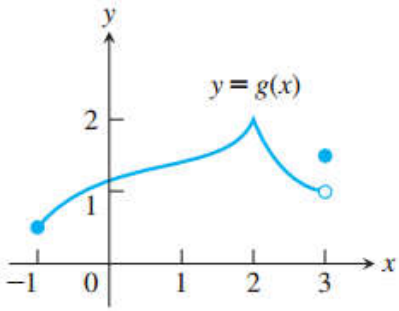
Continuity from Graphs

In Exercises 1–4, say whether the function graphed is continuous on $[-1, 3]$. If not, where does it fail to be continuous and why?

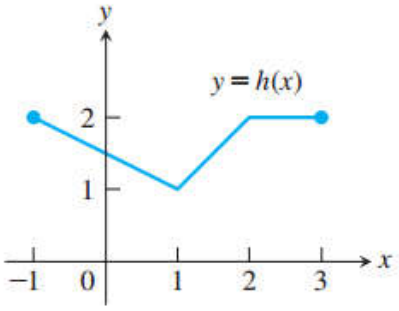
1.



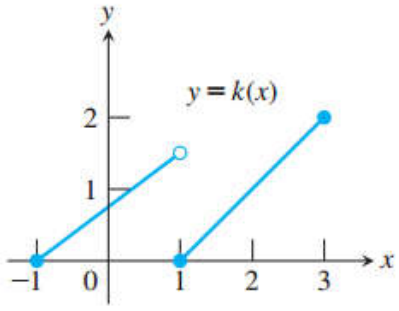
2.



3.



4.



Applying the Continuity Test

At which points do the functions in Exercises 11 and 12 fail to be continuous? At which points, if any, are the discontinuities removable? Not removable? Give reasons for your answers.

11. Exercise 1, Section 2.4

12. Exercise 2, Section 2.4

At what points are the functions in Exercises 13–30 continuous?

13. $y = \dfrac{1}{x-2} - 3x$

14. $y = \dfrac{1}{(x+2)^2} + 4$

15. $y = \dfrac{x+1}{x^2-4x+3}$

16. $y = \dfrac{x+3}{x^2-3x-10}$

17. $y = |x-1| + \sin x$

18. $y = \dfrac{1}{|x|+1} - \dfrac{x^2}{2}$

19. $y = \dfrac{\cos x}{x}$

20. $y = \dfrac{x+2}{\cos x}$

21. $y = \csc 2x$

22. $y = \tan\dfrac{\pi x}{2}$

23. $y = \dfrac{x \tan x}{x^2+1}$

24. $y = \dfrac{\sqrt{x^4+1}}{1+\sin^2 x}$

25. $y = \sqrt{2x+3}$

26. $y = \sqrt[4]{3x-1}$

Continuous Extensions

37. Define $g(3)$ in a way that extends $g(x) = (x^2 - 9)/(x - 3)$ to be continuous at $x = 3$.

38. Define $h(2)$ in a way that extends $h(t) = (t^2 + 3t - 10)/(t - 2)$ to be continuous at $t = 2$.

39. Define $f(1)$ in a way that extends $f(s) = (s^3 - 1)/(s^2 - 1)$ to be continuous at $s = 1$.

40. Define $g(4)$ in a way that extends

$$g(x) = (x^2 - 16)/(x^2 - 3x - 4)$$

to be continuous at $x = 4$.

41. For what value of a is

$$f(x) = \begin{cases} x^2 - 1, & x < 3 \\ 2ax, & x \geq 3 \end{cases}$$

continuous at every x?

42. For what value of b is

$$g(x) = \begin{cases} x, & x < -2 \\ bx^2, & x \geq -2 \end{cases}$$

continuous at every x?

15–20 Explain why the function is discontinuous at the given number a. Sketch the graph of the function.

15. $f(x) = \ln|x - 2|$ $\quad a = 2$

16. $f(x) = \begin{cases} \dfrac{1}{x-1} & \text{if } x \neq 1 \\ 2 & \text{if } x = 1 \end{cases}$ $\quad a = 1$

17. $f(x) = \begin{cases} e^x & \text{if } x < 0 \\ x^2 & \text{if } x \geqslant 0 \end{cases}$ $\quad a = 0$

18. $f(x) = \begin{cases} \dfrac{x^2 - x}{x^2 - 1} & \text{if } x \neq 1 \\ 1 & \text{if } x = 1 \end{cases}$ $\quad a = 1$

19. $f(x) = \begin{cases} \cos x & \text{if } x < 0 \\ 0 & \text{if } x = 0 \\ 1 - x^2 & \text{if } x > 0 \end{cases}$ $\quad a = 0$

20. $f(x) = \begin{cases} \dfrac{2x^2 - 5x - 3}{x - 3} & \text{if } x \neq 3 \\ 6 & \text{if } x = 3 \end{cases}$ $\quad a = 3$

2.6 Limits Involving Infinity; Asymptotes of Graphs

Finite Limits as $x \to \pm\infty$

DEFINITIONS

1. We say that $f(x)$ has the **limit L as x approaches infinity** and write

$$\lim_{x\to\infty} f(x) = L$$

if, for every number $\varepsilon > 0$, there exists a corresponding number M such that for all x in the domain of f

$$|f(x) - L| < \varepsilon \quad \text{whenever} \quad x > M.$$

2. We say that $f(x)$ has the **limit L as x approaches negative infinity** and write

$$\lim_{x\to-\infty} f(x) = L$$

if, for every number $\varepsilon > 0$, there exists a corresponding number N such that for all x in the domain of f

$$|f(x) - L| < \varepsilon \quad \text{whenever} \quad x < N.$$

EXAMPLE 1 Show that

(a) $\lim_{x \to \infty} \frac{1}{x} = 0$ **(b)** $\lim_{x \to -\infty} \frac{1}{x} = 0.$

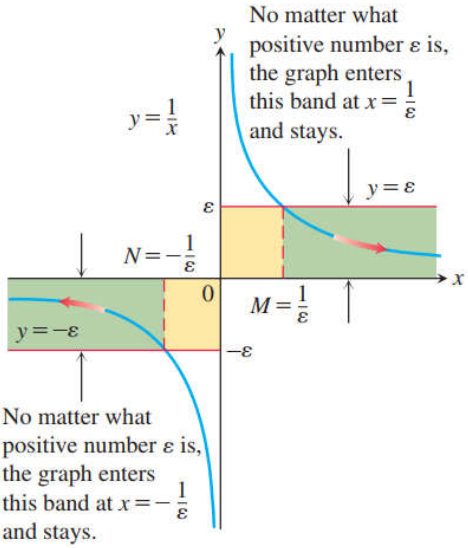


FIGURE 2.50 The geometry behind the argument in Example 1.

THEOREM 12 All the Limit Laws in Theorem 1 are true when we replace $\lim_{x \to c}$ by $\lim_{x \to \infty}$ or $\lim_{x \to -\infty}$. That is, the variable x may approach a finite number c or $\pm\infty$.

DEFINITION A line $y = b$ is a **horizontal asymptote** of the graph of a function $y = f(x)$ if either

$$\lim_{x \to \infty} f(x) = b \qquad \text{or} \qquad \lim_{x \to -\infty} f(x) = b.$$

EXAMPLE 7 Using the Sandwich Theorem, find the horizontal asymptote of the curve

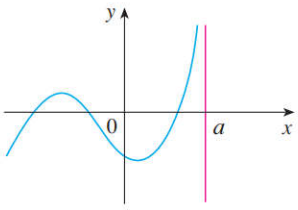
$$y = 2 + \frac{\sin x}{x}.$$

EXAMPLE 8 Find $\lim_{x \to \infty} \left(x - \sqrt{x^2 + 16}\right)$.

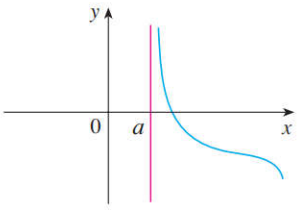
6 DEFINITION The line $x = a$ is called a **vertical asymptote** of the curve $y = f(x)$ if at least one of the following statements is true:

$$\lim_{x \to a} f(x) = \infty \qquad \lim_{x \to a^-} f(x) = \infty \qquad \lim_{x \to a^+} f(x) = \infty$$

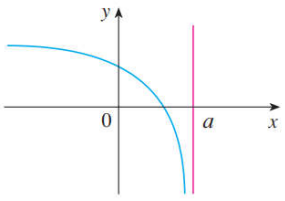
$$\lim_{x \to a} f(x) = -\infty \qquad \lim_{x \to a^-} f(x) = -\infty \qquad \lim_{x \to a^+} f(x) = -\infty$$



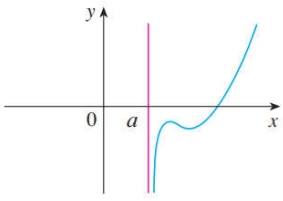
(a) $\lim_{x \to a^-} f(x) = \infty$



(b) $\lim_{x \to a^+} f(x) = \infty$



(c) $\lim_{x \to a^-} f(x) = -\infty$

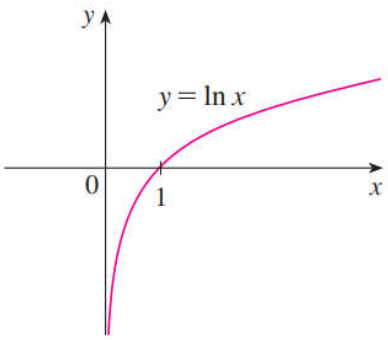


(d) $\lim_{x \to a^+} f(x) = -\infty$

EXAMPLE 9 Find $\lim_{x \to 3^+} \frac{2x}{x - 3}$ and $\lim_{x \to 3^-} \frac{2x}{x - 3}$.

EXAMPLE 10 Find the vertical asymptotes of $f(x) = \tan x$.

Another example of a function whose graph has a vertical asymptote is the natural logarithmic function $y = \ln x$. From Figure 17 we see that



Oblique Asymptotes

If the degree of the numerator of a rational function is 1 greater than the degree of the denominator, the graph has an **oblique** or **slant line asymptote**. We find an equation for the asymptote by dividing numerator by denominator to express f as a linear function plus a remainder that goes to zero as $x \to \pm\infty$.

EXAMPLE 9 Find the oblique asymptote of the graph of

$$f(x) = \frac{x^2 - 3}{2x - 4}$$

in Figure 2.57.

11. Guess the value of the limit

$$\lim_{x \to \infty} \frac{x^2}{2^x}$$

by evaluating the function $f(x) = x^2/2^x$ for $x = 0, 1, 2, 3, 4, 5, 6, 7, 8, 9, 10, 20, 50$, and 100. Then use a graph of f to support your guess.

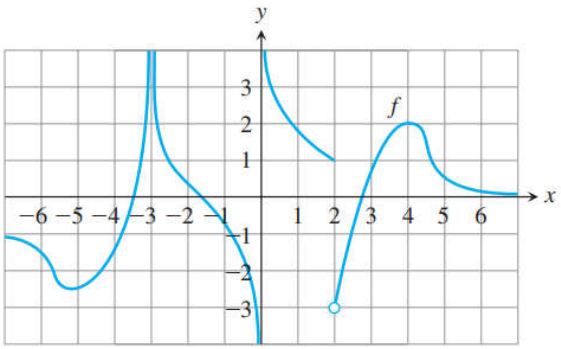
2. For the function f whose graph is given, determine the following limits.

a. $\lim_{x \to 4} f(x)$ **b.** $\lim_{x \to 2^+} f(x)$ **c.** $\lim_{x \to 2^-} f(x)$

d. $\lim_{x \to 2} f(x)$ **e.** $\lim_{x \to -3^+} f(x)$ **f.** $\lim_{x \to -3^-} f(x)$

g. $\lim_{x \to -3} f(x)$ **h.** $\lim_{x \to 0^+} f(x)$ **i.** $\lim_{x \to 0^-} f(x)$

j. $\lim_{x \to 0} f(x)$ **k.** $\lim_{x \to \infty} f(x)$ **l.** $\lim_{x \to -\infty} f(x)$



Find the limits in Exercises 9–12.

9. $\lim_{x\to\infty} \frac{\sin 2x}{x}$

10. $\lim_{\theta\to-\infty} \frac{\cos\theta}{3\theta}$

11. $\lim_{t\to-\infty} \frac{2 - t + \sin t}{t + \cos t}$

12. $\lim_{r\to\infty} \frac{r + \sin r}{2r + 7 - 5\sin r}$

Limits of Rational Functions

In Exercises 13–22, find the limit of each rational function **(a)** as $x \to \infty$ and **(b)** as $x \to -\infty$.

13. $f(x) = \frac{2x + 3}{5x + 7}$

14. $f(x) = \frac{2x^3 + 7}{x^3 - x^2 + x + 7}$

15. $f(x) = \frac{x + 1}{x^2 + 3}$

16. $f(x) = \frac{3x + 7}{x^2 - 2}$

17. $h(x) = \frac{7x^3}{x^3 - 3x^2 + 6x}$

18. $h(x) = \frac{9x^4 + x}{2x^4 + 5x^2 - x + 6}$

19. $g(x) = \frac{10x^5 + x^4 + 31}{x^6}$

20. $g(x) = \frac{x^3 + 7x^2 - 2}{x^2 - x + 1}$

21. $f(x) = \frac{3x^7 + 5x^2 - 1}{6x^3 - 7x + 3}$

22. $h(x) = \frac{5x^8 - 2x^3 + 9}{3 + x - 4x^5}$

Divide numerator and denominator by the highest power of x in the denominator and proceed from there. Find the limits in Exercises 23–36.

23. $\lim_{x\to\infty} \sqrt{\dfrac{8x^2-3}{2x^2+x}}$

24. $\lim_{x\to-\infty} \left(\dfrac{x^2+x-1}{8x^2-3}\right)^{1/3}$

25. $\lim_{x\to-\infty} \left(\dfrac{1-x^3}{x^2+7x}\right)^5$

26. $\lim_{x\to\infty} \sqrt{\dfrac{x^2-5x}{x^3+x-2}}$

27. $\lim_{x\to\infty} \dfrac{2\sqrt{x}+x^{-1}}{3x-7}$

28. $\lim_{x\to\infty} \dfrac{2+\sqrt{x}}{2-\sqrt{x}}$

29. $\lim_{x\to-\infty} \dfrac{\sqrt[3]{x}-\sqrt[5]{x}}{\sqrt[3]{x}+\sqrt[5]{x}}$

30. $\lim_{x\to\infty} \dfrac{x^{-1}+x^{-4}}{x^{-2}-x^{-3}}$

31. $\lim_{x\to\infty} \dfrac{2x^{5/3}-x^{1/3}+7}{x^{8/5}+3x+\sqrt{x}}$

32. $\lim_{x\to-\infty} \dfrac{\sqrt[3]{x}-5x+3}{2x+x^{2/3}-4}$

33. $\lim_{x\to\infty} \dfrac{\sqrt{x^2+1}}{x+1}$

34. $\lim_{x\to-\infty} \dfrac{\sqrt{x^2+1}}{x+1}$

35. $\lim_{x\to\infty} \dfrac{x-3}{\sqrt{4x^2+25}}$

36. $\lim_{x\to-\infty} \dfrac{4-3x^3}{\sqrt{x^6+9}}$

Infinite Limits

Find the limits in Exercises 37–48.

37. $\lim_{x \to 0^+} \frac{1}{3x}$

38. $\lim_{x \to 0^-} \frac{5}{2x}$

39. $\lim_{x \to 2^-} \frac{3}{x - 2}$

40. $\lim_{x \to 3^+} \frac{1}{x - 3}$

41. $\lim_{x \to -8^+} \frac{2x}{x + 8}$

42. $\lim_{x \to -5^-} \frac{3x}{2x + 10}$

43. $\lim_{x \to 7} \frac{4}{(x - 7)^2}$

44. $\lim_{x \to 0} \frac{-1}{x^2(x + 1)}$

45. a. $\lim_{x \to 0^+} \frac{2}{3x^{1/3}}$ **b.** $\lim_{x \to 0^-} \frac{2}{3x^{1/3}}$

46. a. $\lim_{x \to 0^+} \frac{2}{x^{1/5}}$ **b.** $\lim_{x \to 0^-} \frac{2}{x^{1/5}}$

47. $\lim_{x \to 0} \frac{4}{x^{2/5}}$

48. $\lim_{x \to 0} \frac{1}{x^{2/3}}$

Graphing Simple Rational Functions

Graph the rational functions in Exercises 63–68. Include the graphs and equations of the asymptotes and dominant terms.

63. $y = \dfrac{1}{x - 1}$

64. $y = \dfrac{1}{x + 1}$

65. $y = \dfrac{1}{2x + 4}$

66. $y = \dfrac{-3}{x - 3}$

67. $y = \dfrac{x + 3}{x + 2}$

68. $y = \dfrac{2x}{x + 1}$

Domains, Ranges, and Asymptotes

Determine the domain and range of each function. Use various limits to find the asymptotes and the ranges.

69. $y = 4 + \dfrac{3x^2}{x^2 + 1}$

70. $y = \dfrac{2x}{x^2 - 1}$

71. $y = \dfrac{\sqrt{x^2 + 4}}{x}$

72. $y = \dfrac{x^3}{x^3 - 8}$

Inventing Graphs and Functions

In Exercises 73–76, sketch the graph of a function $y = f(x)$ that satisfies the given conditions. No formulas are required—just label the coordinate axes and sketch an appropriate graph. (The answers are not unique, so your graphs may not be exactly like those in the answer section.)

73. $f(0) = 0, f(1) = 2, f(-1) = -2, \lim_{x \to -\infty} f(x) = -1$, and $\lim_{x \to \infty} f(x) = 1$

74. $f(0) = 0, \lim_{x \to \pm\infty} f(x) = 0, \lim_{x \to 0^+} f(x) = 2$, and $\lim_{x \to 0^-} f(x) = -2$

75. $f(0) = 0, \lim_{x \to \pm\infty} f(x) = 0, \lim_{x \to 1^-} f(x) = \lim_{x \to -1^+} f(x) = \infty$, $\lim_{x \to 1^+} f(x) = -\infty$, and $\lim_{x \to -1^-} f(x) = -\infty$

76. $f(2) = 1, f(-1) = 0, \lim_{x \to \infty} f(x) = 0, \lim_{x \to 0^+} f(x) = \infty$, $\lim_{x \to 0^-} f(x) = -\infty$, and $\lim_{x \to -\infty} f(x) = 1$

Finding Limits of Differences When $x \to \pm\infty$

Find the limits in Exercises 84–90. (*Hint*: Try multiplying and dividing by the conjugate.)

84. $\lim_{x\to\infty} (\sqrt{x+9} - \sqrt{x+4})$

85. $\lim_{x\to\infty} (\sqrt{x^2+25} - \sqrt{x^2-1})$

86. $\lim_{x\to-\infty} (\sqrt{x^2+3} + x)$

87. $\lim_{x\to-\infty} (2x + \sqrt{4x^2+3x-2})$

88. $\lim_{x\to\infty} (\sqrt{9x^2-x} - 3x)$

89. $\lim_{x\to\infty} (\sqrt{x^2+3x} - \sqrt{x^2-2x})$

90. $\lim_{x\to\infty} (\sqrt{x^2+x} - \sqrt{x^2-x})$

Oblique Asymptotes

Graph the rational functions in Exercises 103–108. Include the graphs and equations of the asymptotes.

103. $y = \dfrac{x^2}{x - 1}$

104. $y = \dfrac{x^2 + 1}{x - 1}$

105. $y = \dfrac{x^2 - 4}{x - 1}$

106. $y = \dfrac{x^2 - 1}{2x + 4}$

107. $y = \dfrac{x^2 - 1}{x}$

108. $y = \dfrac{x^3 + 1}{x^2}$

Horizontal and Vertical Asymptotes

47. Use limits to determine the equations for all vertical asymptotes.

a. $y = \dfrac{x^2 + 4}{x - 3}$ **b.** $f(x) = \dfrac{x^2 - x - 2}{x^2 - 2x + 1}$

c. $y = \dfrac{x^2 + x - 6}{x^2 + 2x - 8}$

48. Use limits to determine the equations for all horizontal asymptotes.

a. $y = \dfrac{1 - x^2}{x^2 + 1}$ **b.** $f(x) = \dfrac{\sqrt{x} + 4}{\sqrt{x + 4}}$

c. $g(x) = \dfrac{\sqrt{x^2 + 4}}{x}$ **d.** $y = \sqrt{\dfrac{x^2 + 9}{9x^2 + 1}}$

49. Determine the domain and range of $y = \dfrac{\sqrt{16 - x^2}}{x - 2}$.

50. Assume that constants a and b are positive. Find equations for all horizontal and vertical asymptotes for the graph of $y = \dfrac{\sqrt{ax^2 + 4}}{x - b}$.